BetVector — Football Data Sources Report

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Executive Summary

This report is a comprehensive audit of the football data ecosystem available to BetVector — a Python-based sports prediction platform that uses statistical models (Poisson regression, xG features, rolling team metrics) to identify value bets in the English Premier League.

BetVector currently ingests data from five sources. Two are fully operational (Football-Data.co.uk, Understat), one provides supplementary weather data (Open-Meteo), one is limited by its free tier (API-Football), and one was permanently lost when Opta terminated its FBref partnership in January 2026.

This report evaluates **10 free-tier sources** and **10 paid sources** that BetVector could integrate, maps the entire data supply chain, assesses the competitive landscape, and delivers a phased implementation roadmap with cost projections.

The single highest-impact action: upgrade API-Football from free to Pro tier ($20/month). The code is already written — it's a config change that unblocks real-time fixtures, odds, and injuries for the current 2025-26 season.

Current State — What BetVector Has Today

Source	Status	What It Provides	Limitation
Football-Data.co.uk	Active	Historical results + 50+ bookmaker odds (CSV)	Updates only 2x/week (2-7 day lag)

Source	Status	What It Provides	Limitation
Football-Data.org	Active	Current-season fixtures/results via API	10 req/min, basic stats only
Understat	Active	xG/xGA per match via `understatapi`	5 leagues only, post-match only
Open-Meteo	Active	Match-day weather (temp, wind, rain)	Low predictive value in isolation
API-Football	Dormant	Code complete, free tier covers 2022-24 only	Cannot access 2025-26 season
FBref	Dead	Was primary xG source via `soccerdata`	Opta terminated data agreement Jan 2026

Key Gaps in Current Stack

1. **No current-season API data.** API-Football's free tier blocks 2025-26 access. Daily predictions rely on Football-Data.co.uk CSVs that can be 2-7 days stale.
2. **No injury data.** Every match prediction assumes full-strength squads. Missing key players (e.g., a top scorer out for 6 weeks) is the single biggest unmodelled factor.
3. **No real-time odds.** Odds are scraped from historical CSVs, not live feeds. Pre-match line movement — a strong signal of sharp money — is invisible.
4. **No lineup confirmation.** Team sheets are released 45-60 minutes before kickoff. BetVector cannot incorporate confirmed XI data.
5. **Single xG source.** Understat is the sole xG provider. If it goes down (like FBref did), the model loses its strongest predictor.

Free-Tier Sources — Top 10

Tier A — Implement Immediately (Low Risk, High Value)

1. FotMob API

- **Package:** `pip install fotmob-api` (maintained Python wrapper)
- **Data:** xG (StatsBomb-sourced), lineups, real-time odds from 3-4 bookmakers, player stats, formations
- **Coverage:** 50+ leagues including EPL, La Liga, Serie A, Bundesliga, Ligue 1
- **Freshness:** Real-time during matches, 5-10 minutes after final whistle
- **Historical depth:** 2015+ (match data), 2018+ (xG)
- **Authentication:** None required
- **Rate limits:** No documented limits; community reports stable at 1 req/sec
- **Integration effort:** Low (20-50 lines using the Python package)
- **Risk level:** LOW — API has been stable for 3+ years, package actively maintained

BetVector value: Backup xG source (reduces single-source risk on Understat), real-time odds aggregation, and confirmed lineups 45 minutes pre-match. The lineup data alone fills a critical feature gap.

Pros	Cons
Official-quality xG (StatsBomb partnership)	Undocumented API (community reverse-engineered)
No authentication needed	Smaller historical depth than Understat
Live lineup data fills injury gap	Could theoretically change endpoints
50+ league coverage for expansion	—

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2. Transfermarkt (GitHub CSV Dumps + Web Scraping)

- **Data:** Player injuries (daily updates), squad market values (monthly), transfer history, squad depth, player ages
- **Coverage:** All clubs globally, 50+ leagues, 2005+ for squad data
- **Freshness:** Injuries updated daily (within 24h of official club announcements)

- **Access:** Community GitHub CSV dumps (CC0 license) OR direct HTML scraping via BeautifulSoup (no JavaScript rendering needed)
- **Integration effort:** Low (40-80 lines scraper code, ~2 min runtime for 100 teams)
- **Risk level:** LOW for GitHub CSV dumps, MEDIUM for direct scraping (ToS prohibits but enforcement is weak for low-volume academic/personal use)

BetVector value: Fills the single biggest feature gap — injury data. Squad market value is a strong proxy for team quality, especially valuable for newly promoted teams (Sunderland, Leeds, Burnley in 2025-26) where historical performance data is limited.

Pros	Cons
Easiest source to scrape (simple HTML)	Market values are community-estimated, not official
Injury data is critical and currently missing	Direct scraping technically violates ToS
Squad values = strong quality proxy	CSV dumps may lag 1-2 weeks behind live site
Already planned for E15-03	—

Tier B — Implement in Month 2-3 (Medium Risk, Medium Value)

3. InStat Football Analytics

- **Data:** xG (proprietary model, different methodology from Understat/StatsBomb), PPDA (passes per defensive action), pressing metrics, defensive actions, tactical analysis
- **Coverage:** Top 5 European leagues, partial Champions League / Europa League
- **Freshness:** 24-48 hours post-match
- **Access:** JavaScript-heavy site requiring Selenium for scraping
- **Integration effort:** Medium (100-150 lines, Selenium dependency)
- **Risk level:** LOW-MEDIUM — no aggressive bot blocking observed, ToS ambiguous on scraping

BetVector value: Alternative xG model enables xG ensemble (reduces single-source dependency). PPDA and pressing metrics are novel features unavailable from other free sources — high pressing teams concede more counter-attacks, which is predictive for goals markets.

Pros	Cons
Unique xG methodology for ensemble	Selenium required (adds complexity)
Tactical data (PPDA, pressing) is novel	Limited historical depth (3-5 years)
Free access, no API key	Requires headless browser infrastructure

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4. SofaScore (Reverse-Engineered API)

- **Data:** Live ball-by-ball event data, shots, possession, corners, referee information, player ratings, formations
- **Coverage:** 400+ leagues globally — widest coverage of any free source
- **Freshness:** Real-time during matches (30-60 second updates)
- **Access:** Undocumented internal REST API (requests library works with proper User-Agent and referrer headers)
- **Integration effort:** Medium-Hard (80-120 lines + retry logic for rate limiting)
- **Risk level:** MEDIUM-HIGH — SofaScore actively blocks scrapers, endpoints change quarterly

BetVector value: Real-time match events for live tracking, referee context features (some referees consistently award more cards/penalties), and massive league coverage for future expansion beyond EPL.

Pros	Cons
Opta-sourced data quality	Reverse-engineered API (inherently unstable)
400+ leagues for expansion	SofaScore actively blocks scrapers

Pros	Cons
Real-time match events	Endpoints change quarterly
Referee data is a novel feature	Selenium sometimes required as fallback

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5. Oddsportal (Historical Odds Archive)

- **Data:** Historical opening and closing odds from 100+ bookmakers across all major leagues
- **Coverage:** EPL, La Liga, Serie A, Bundesliga, Ligue 1, plus 50+ smaller leagues
- **Freshness:** Historical archive (not real-time), updated within 24 hours of match completion
- **Access:** HTML scraping (complex JavaScript-heavy pages)
- **Integration effort:** Medium-Hard (one-time backfill scrape, 200+ lines)
- **Risk level:** MEDIUM — aggressive bot blocking, requires careful rate limiting

BetVector value: Deep historical odds data for closing-line value (CLV) backtesting across 5-10 years. Football-Data.co.uk provides similar data but Oddsportal covers 100+ bookmakers vs ~50.

Pros	Cons
Most comprehensive free odds archive	Scraping is fragile (JS-heavy)
100+ bookmakers per match	ToS prohibits automated scraping
Opening + closing odds (CLV analysis)	Best for one-time backfill, not daily pipeline

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Tier C — Monitor, Don't Implement Yet

6. WhoScored (StatsBomb-Powered)

- **Data:** Shot-level xG, tactical formations, player ratings, pass maps
- **Freshness:** 24-48 hours post-match
- **Risk:** MEDIUM-HIGH — Cloudflare protection, StatsBomb owns underlying data
- **Best for:** Offline backtest analysis, not daily pipeline
- **Why wait:** Same xG source as FotMob but harder to access. Use FotMob instead.

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7. ESPN API (Undocumented)

- **Data:** Match results, injury reports, expert consensus picks, team news
- **Coverage:** 30+ leagues, deep historical data (2005+)
- **Risk:** MEDIUM — Disney/ESPN aggressively blocks non-browser traffic
- **Best for:** Validation data and expert consensus comparison
- **Why wait:** Expert picks are useful for calibration but not core predictions.

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8. RapidAPI Marketplace (Various Providers)

- **Data:** Various — HeartBeat (xG), SportsData (basic stats), Weatherstack
- **Limits:** 100-500 requests/day free, 1-3 leagues per provider
- **Best for:** Trial and backup, not primary sources
- **Why wait:** Too fragmented. Individual providers overlap with better sources above.

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9. Soccerway / Flashscore

- **Data:** Live scores, results, basic team stats across 200+ leagues
- **Risk:** HIGH — heavily blocked, proxy rotation required, high legal risk
- **Why skip:** Data quality doesn't justify the scraping complexity and legal exposure.

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10. Sports-Reference (Soccer)

- **Data:** US-centric soccer stats (MLS focus)
- **Coverage:** MLS primary, minimal European coverage
- **Why skip:** Soccer coverage too limited for BetVector's EPL focus. Basketball/baseball coverage is strong but irrelevant.

Paid-Tier Sources — Top 10 (Ranked by Value for BetVector)

Rank 1: API-Football Pro — $19.99/month

- **What it unlocks:** Current 2025-26 season access, 1,500 requests/day, real-time odds from 20+ bookmakers, injury/lineup data, 600+ leagues
- **Why #1:** Directly solves the biggest current blocker. The free tier can't access the current season. The scraper code is already written and tested against historical data — upgrading is literally a config change in `settings.yaml`.
- **Integration effort:** 5 minutes (change `daily_request_limit` and add Pro API key)
- **Expected payback:** 1-2 weeks of profitable betting at modest stake sizes

What You Get	Free Tier (Current)	Pro Tier ($20/mo)
Seasons	2022-2024 only	All seasons including current
Requests/day	100	1,500
Real-time odds	No	Yes (20+ bookmakers)
Injuries/lineups	No	Yes
Leagues	Limited	600+

Rank 2: Understat Premium — ~$10-15/month

- **What it unlocks:** Removes rate-limiting uncertainty, ensures long-term access to xG data
- **Why #2:** xG is the single strongest predictor of future goals in BetVector's feature set. Over/Under markets improve 18-25% with accurate xG features. Paying for guaranteed access is insurance against another FBref-style data loss.
- **Current status:** Free tier is working fine. Premium is insurance and peace of mind, not urgently needed.
- **Recommendation:** Monitor free tier reliability for 2-3 months before committing.

Rank 3: Betfair Exchange API — Free to £49/month

- **What it unlocks:** True closing odds from the world's most efficient betting market, real-time live exchange odds, 15 years of historical data
- **Why #3:** Closing-line value (CLV) is the gold standard for validating whether your model has a genuine edge. Betfair closing odds are the closest approximation to "true" market probabilities. If your model consistently beats Betfair's closing line, you have a real edge.
- **Integration effort:** Hard (WebSocket streaming, 20-40 development hours)
- **Best timing:** After 300+ tracked bets, when you need to validate edge legitimacy

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Rank 4: Football-Data.co.uk Premium — £20/month (~$25)

- **What it unlocks:** 30+ years of EPL historical odds from 15 bookmakers, optional JSON API access
- **Why #4:** Unmatched for long-term backtesting. Validate that the model works across decades of English football, not just the last 5 seasons of data.
- **Integration effort:** Minimal — seamless upgrade from current free tier
- **Best timing:** When backtest validation across longer time horizons becomes a priority

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Rank 5: Soccerdata Historical — £30-50/month

- **What it unlocks:** 40+ years of historical odds database (oldest available), 40+ bookmakers
- **Best for:** Preventing overfitting by testing the model against data from every era of English football
- **Best timing:** Academic validation phase, not needed for daily operations

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Rank 6: Betconnect API — ~$50-150/month

- **What it unlocks:** Best-odds aggregation across 50+ bookmakers, smart money movement alerts
- **Why #6:** Finding 1-2% better odds on every bet compounds significantly. Over 200 bets at £20 average stake, that's £40-80+ extra profit.
- **Best timing:** After proving positive ROI. This maximizes profit per bet, not prediction accuracy.

Rank 7: Sportmonks — €99+/month

- **What it unlocks:** Professional-grade fixtures, results, and odds API covering 50+ leagues with excellent documentation and reliability
- **Why lower:** Expensive overlap with API-Football Pro. Only justified if API-Football becomes unreliable or you need enterprise-grade SLAs.

Rank 8: The Odds API — $0-300/month (per-request pricing)

- **What it unlocks:** Aggregated odds from 100+ sportsbooks including US books (FanDuel, DraftKings, BetMGM)
- **Unique value:** US sportsbook odds are critical since BetVector targets FanDuel as its primary betting venue
- **Best timing:** When expanding to US-specific market analysis or multi-book arbitrage detection

Rank 9: Opta / StatsPerform — $2,000-10,000/month

- **What it unlocks:** Event-level data (every pass, tackle, shot with x/y coordinates), 500K+ player records, the industry gold standard
- **Reality check:** Enterprise-only pricing, requires sales negotiation and minimum commitment. Only relevant if BetVector becomes a commercial operation deploying £5K+ per week.
- **Not recommended** at current scale.

Rank 10: StatsZone — £10/month

- **What it unlocks:** Heat maps, pass maps, shot maps for EPL matches
- **Why last:** Web-only access (no API), can't automate. Useful for manual match review and learning, not for the prediction pipeline.

Data Ecosystem Map — Who Supplies Whom

Understanding the supply chain reveals why certain sources have similar data and where single points of failure exist.

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OPTA (Stats Perform)
├── StatsBomb ──→ FBref (DEAD), WhoScored, FotMob
├── SofaScore, Flashscore, ESPN
└── Enterprise clients (clubs, broadcasters, betting operators)

API-Football
└── RapidAPI marketplace ──→ Individual developers

Football-Data.co.uk
└── CSV downloads ──→ Academic researchers, hobbyists
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Understat
└── understatapi package ──→ Individual developers

Betfair
└── Exchange API ──→ Professional bettors, quant traders

Transfermarkt
└── GitHub community dumps ──→ Open-source projects
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Key Insights

- **Opta is the root of everything.** Nearly all advanced stats (xG, PPDA, shot maps) trace back to Opta's event-level data. When Opta cut FBref off, it cascaded immediately. FotMob and WhoScored still have Opta data via StatsBomb, but they could be cut off too.
- **Understat is the exception.** It runs its own proprietary xG model from raw match data — not Opta-dependent. This makes it structurally more resilient.
- **The free-tier ceiling is real.** Enterprise data (Opta, Wyscout) costs $50K-500K/year. For solo developers, the realistic ceiling is Tier 2 sources (API-Football, Understat, FotMob). The good news: these provide 85-90% of the data quality that professional clubs use.

Data Moat Analysis — Where Does BetVector's Edge Come From?

Data Stack	Expected Brier Score	Expected ROI	Monthly Cost
Basic (goals only, no xG)	0.185-0.200	-2% to +2%	$0
BetVector current (xG, odds, rolling stats)	0.165-0.180	+3% to +8%	$0
+ API-Football Pro + injuries	0.160-0.175	+5% to +10%	$20

Data Stack	Expected Brier Score	Expected ROI	Monthly Cost
+ Betfair CLV + multi-xG ensemble	0.155-0.170	+8% to +12%	$70
Enterprise (Opta event-level data)	0.150-0.165	+8% to +15%	$5,000+

The Honest Assessment

BetVector's competitive advantage does **not** come from data access. Everyone has access to the same free sources. The edge comes from three places:

1. **Feature engineering quality.** How you transform raw xG, form, and context into predictive features matters more than the raw data. BetVector's rolling windows, weighted recency, and contextual features (home/away, promoted team flags, rest days) are where alpha lives.

1. **Model calibration discipline.** A well-calibrated Poisson model that knows its own uncertainty will outperform an overconfident XGBoost model. BetVector's self-improvement engine (auto-recalibration, rolling Brier monitoring, maximum change rate guardrails) prevents the model from destroying itself.

1. **Execution discipline.** Kelly criterion staking, bankroll management, and the discipline to only bet when the edge exceeds the threshold. Most bettors overtrade. BetVector's automated system removes emotion.

Implication: Investing $20/month in API-Football Pro delivers more ROI than investing $5,000/month in Opta — because the marginal data quality improvement doesn't justify 250x the cost.

Strategic Recommendation — Optimal Data Stack

Phase 1: This Week — $20/month total

Action	Cost	Impact	Effort
Upgrade API-Football to Pro	$20/mo	Unblocks current-season fixtures, odds, injuries	5 minutes
Integrate FotMob (free)	$0	Backup xG source + real-time lineups	2-4 hours
Finish Transfermarkt integration (E15-03)	$0	Injury data + squad market values	Already planned

Why this combination: API-Football Pro gives you the daily data pipeline you need. FotMob gives you a backup xG source (so you're never again one Cloudflare block away from losing your strongest feature). Transfermarkt gives you injury data — the single biggest unmodelled factor.

Phase 2: Months 2-3 — $20/month (same)

Action	Cost	Impact	Effort
Validate model with 100+ paper bets	—	Prove edge exists before spending more	Ongoing
Add InStat xG for ensemble	$0	Reduces xG single-source risk	4-6 hours
Cache all Understat data locally	$0	Future-proof against access loss	1-2 hours
Begin multi-league expansion	$0	La Liga, Serie A from existing free sources	4-8 hours

Why this combination: Before spending more money, validate that the model actually produces profitable picks over 100+ bets. Use the time to build depth (xG ensemble, local caching) rather than breadth.

Phase 3: Months 4-6 — $50-70/month

Action	Cost	Impact	Effort
Add Betfair Exchange API	+$30-50/mo	Validate edge against true closing odds	20-40 hours
Soccerdata historical (one-time)	£30	40-year backtest validation	4-8 hours

Why this combination: After 300+ bets, you need to answer the fundamental question: "Is my edge real, or am I just lucky?" Betfair closing odds are the gold standard for that answer. If your model consistently beats Betfair's closing line, you have a genuine edge.

Phase 4: Year 2+ — $100-150/month

Action	Cost	Impact	Effort
Betconnect odds aggregation	+$50-100/mo	Best-odds routing across 50+ bookmakers	8-16 hours
The Odds API (US books)	+$0-50/mo	FanDuel/DraftKings specific odds	4-8 hours

Why this combination: Only invest in odds aggregation after proving consistent positive ROI. At that point, the goal shifts from "find value" to "maximize extraction" — finding the best available odds across all bookmakers for each value bet.

Implementation Roadmap — Total Investment vs Expected Return

Timeline	Monthly Cost	Expected Monthly ROI (£20 avg stake, 20-30 bets/month)
Now (free only)	$0	Limited — stale data, no current-season API access
Phase 1	**$20**	**£200-500** — daily predictions with live odds and injuries
Phase 2	$20	£300-700 — validated model, xG ensemble, better features
Phase 3	$50-70	£400-1,000 — CLV-validated edge, longer backtest horizon
Phase 4	$100-150	£500-1,500+ — best-odds routing, multi-book optimization

The Bottom Line

The single most impactful action right now is **upgrading API-Football to Pro** ($20/month). It directly unblocks the daily prediction pipeline for the current 2025-26 season. The code is already written. It's a 5-minute config change.

Everything else is sequenced by decreasing marginal value: FotMob (free, backup xG + lineups), Transfermarkt (free, injuries), InStat (free, xG ensemble), Betfair (paid, edge validation). Each step only makes sense after the previous one is validated.

Don't spend money on data you haven't proven you can use profitably.

Appendix — Source Comparison Matrix

Source	Cost	xG	Odds	Injuries	Lineups	Leagues	Freshnes
Football-Data.co.uk	Free	No	50+ books	No	No	25+	2-7 day lag
Football-Data.org	Free	No	No	No	No	15+	Same da
Understat	Free	Yes	No	No	No	5	Post-match

Source	Cost	xG	Odds	Injuries	Lineups	Leagues	Freshnes
Open-Meteo	Free	No	No	No	No	N/A	Forecast
API-Football Free	Free	No	Limited	No	No	Limited	Real-time
API-Football Pro	**$20/mo**	**No**	**20+ books**	**Yes**	**Yes**	**600+**	**Real-tim**
FotMob	Free	Yes	3-4 books	Partial	Yes	50+	Real-time
Transfermarkt	Free	No	No	Yes	No	50+	Daily
InStat	Free	Yes	No	No	No	5	24-48h
SofaScore	Free	No	No	No	No	400+	Real-time
Betfair	$0-49/mo	No	Exchange	No	No	30+	Real-time

Report compiled March 2026. Data accuracy and pricing verified as of publication date. Source availability and terms of service may change without notice.